

Exercise 7 Transfer-Control and Internet Protocol

Learning Objectives:

- Name the TCP/IP Reference model including all layers
- Distinguish the functions of the layers in the TCP/IP model
- Explain typical protocols of the layers in the TCP/IP model

Legend:

 Reading that must be completed before the exercise

 Questions/tasks to be worked on before the exercise

 Tasks to be worked on during the exercise

7.1 Preparations for the Exercise

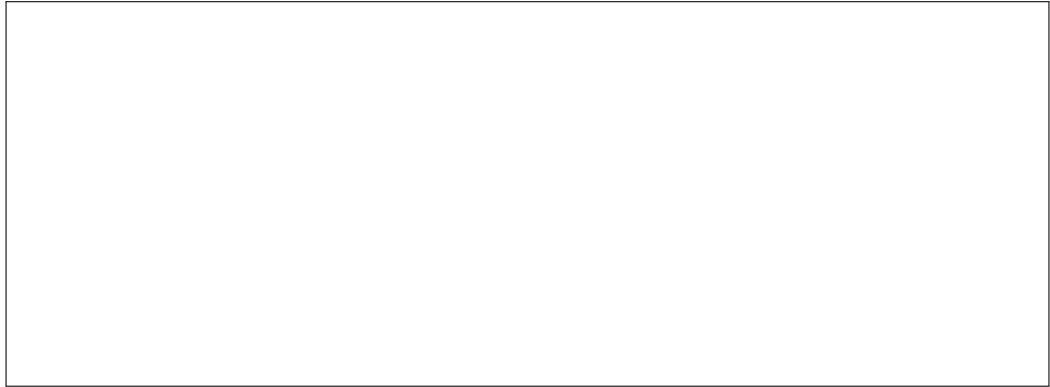
The preparations and questions must be completed before the exercise session. The exception is this first exercise, as you have not yet had the opportunity to read the material. These are also intended as aids to make the tasks that need to be worked on during the exercise session easier for you.

 As preparation, study the following materials, at least one of the videos:

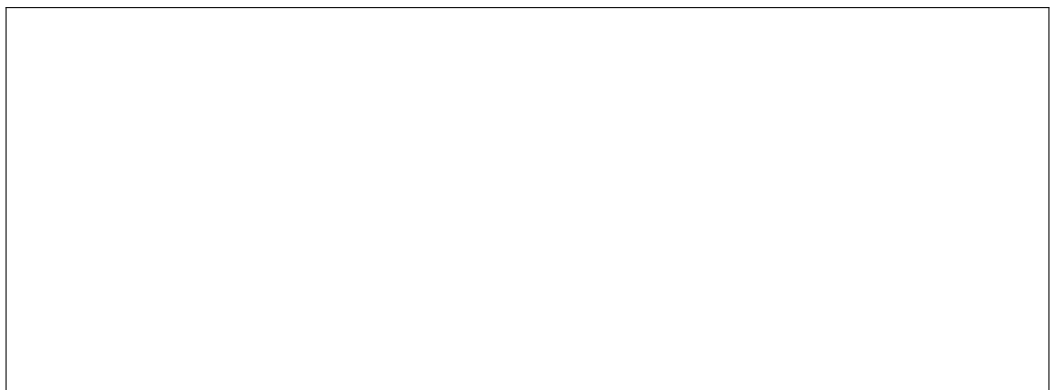
- YT - Video (11min): Learn Wireshark in 10 minutes - Wireshark Tutorial for Beginners
- YT - Video (55min): Mastering Wireshark: The Complete Tutorial!
- StationX: How to Use Wireshark to Capture Network Traffic (2024)
- Christian Baun, *Computer Networks / Computernetze*, Bilingual 2nd Edition,

 **Introductory Questions:**

- a) Draw the TCP/IP layer model.



- b) Provide at least two protocol examples for each of the four TCP/IP layers.



7.2 Introduction to Wireshark

🖱️ Install Wireshark on your own system. Additional information is given by the video references on top of this exercise. ❖ Start Wireshark on your

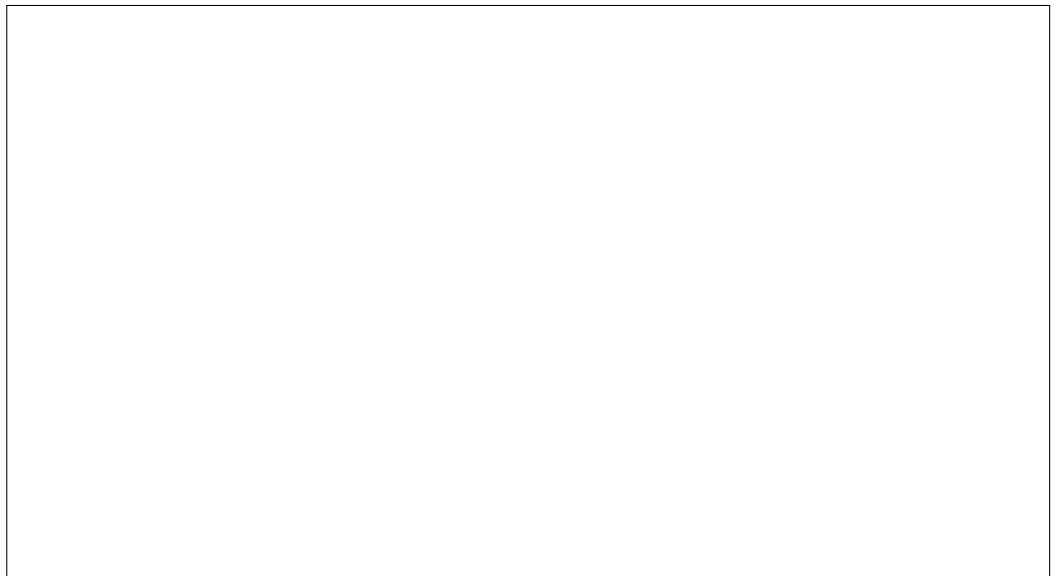
virtual or native Linux system (or Windows system, if you have installed Wireshark there).

- a) Start a capture session for the active network interface in Wireshark. This interface depends on your device's connection to the Internet, usually WLAN or LAN.
- b) Start a web browser parallel to Wireshark and access the URL <http://www.th-ab.de>. Please note, that you ask for an **http** without **s** connection.
- c) Stop the capture session with Wireshark and analyze the data in Wireshark so that you:
 - (a) Identify the various protocols necessary for contacting the web server up to displaying the website in the web browser. Write/Copy these in the order they appear.

- (b) Locate the first occurrence of an HTTPS request. Note the structure of the vertical communication in terms of data packet, trailer, and header of the respective layer. Describe the number of header and trailer bytes.



- (c) Determine the MAC address of your device based on the communication flow. What is this client MAC address?



- (d) Similarly, identify the MAC address of the server from the communication flow.

